Decoupling acceptance from scaling

what it costs to accept a step the radius does not trust

Abstract

The three thresholds of a trust-region method answer different questions. The acceptance threshold η decides whether a step is taken. The scaling thresholds η_1 and η_2 decide whether the region shrinks or grows. Setting $\eta < \eta_1$ opens a band on which a step is accepted and the radius still contracts. We sweep η from 0 to η_1 for eight radius rules on 185 unconstrained CUTEst problems, at two values of η_1, and measure the occupancy of that band before measuring anything else. PLACEHOLDER-ABSTRACT

1 What Part I says about $\eta = 0$

The question decides what the rest of this report means. Either the structural conditions on $\{\Delta_k\}$ supply the quantified decrease that the classical complexity argument takes from η, or the $\mathcal{O}(\varepsilon^{-2})$ bound does not survive $\eta = 0$. We read Part I to settle it. Every label below is quoted from that source.

Part I fixes $0 \le \eta \le \eta_1 \le \eta_2 < 1$ and defines $\mathcal{S} = \{k \ge 0 : \rho_k \ge \eta\}$ and $\mathcal{U} = \{k \ge 0 : \rho_k < \eta\}$. The abstract states the extension directly. Part I revisits the framework of Curtis and Scheinberg and of Wang and Yuan, and extends it to three distinct scaling factors with decoupled step acceptance, covering $\eta = 0$. The body attributes the Lyapunov framework to the survey of Curtis and Scheinberg as developed by Cartis and Scheinberg and by Blanchet, Cartis, Menickelly and Scheinberg, and records that the extension allows $\eta = 0$, which is not treated there.

Part I also states the classical position plainly. Several authors have used $\eta = 0$, accepting any step that decreases the objective. That choice ensures only that at least one accumulation point is first-order critical, which is $\liminf_{k\to\infty} \|g_k\| = 0$. Part I cites Yuan's one-dimensional example, which cycles among three points, two of them non-stationary.

The decrease is quantified by a different threshold. Three conditions carry the analysis. Condition C.LbR bounds the radius from below. Condition C.DecR forces contraction after a poor step. Condition C.Exp/DecR caps expansion after a good one.

Condition C.LbR `cond: delta_k lower bound`. There is $0 < \kappa_{\text{lbd}} \le 1$ with $\Delta_k \ge \frac{\kappa_{\text{lbd}}}{M_k} \min_{0 \le i \le k} \|g_i\|$ for all k, where $\{M_k\}$ is non-decreasing and bounds the model Hessian norm.

Condition C.DecR `cond: decrease of the trust-region radius`. There are $\underline{\gamma_2} \in (0,1)$ and $\eta_{\text{dec}} > 0$ such that $\rho_k < \eta_{\text{dec}}$ implies $\Delta_{k+1} \le \underline{\gamma_2}\,\Delta_k$.

Condition C.Exp/DecR `cond: increase of the trust-region radius`. There are $\underline{\gamma_2} \in (0,1)$, $\overline{\gamma_3} > 1$ and $\eta_{\text{inc}} > 0$ such that $\rho_k < \eta_{\text{inc}}$ implies $\Delta_{k+1} \le \underline{\gamma_2}\,\Delta_k$, and $\rho_k \ge \eta_{\text{inc}}$ implies $\Delta_{k+1} \le \overline{\gamma_3}\,\Delta_k$.

The thresholds η_{dec} and η_{inc} belong to the radius rule. They are required to be strictly positive by the conditions themselves. They are not η.

This is the whole answer. Part I's liminf result, `thm: Liminf of the gradient norm under strong assumption`, assumes Condition C.LbR and Condition C.DecR with $\eta_{\text{dec}} \geq \eta$ and $\eta_{\text{dec}} \neq 0$, and concludes $\liminf_{k\to\infty} \|g_k\| = 0$. At $\eta = 0$ the requirement $\eta_{\text{dec}} \geq \eta$ is satisfied by every admissible rule, and $\eta_{\text{dec}} \neq 0$ is a property of the rule rather than of the acceptance test. The liminf result therefore holds at $\eta = 0$. The companion result under the weaker Hessian assumption, `thm: Liminf of the gradient norm under weaker assumption`, is stated the same way with $\eta_{\text{inc}} \geq \eta$ and $\eta_{\text{inc}} \neq 0$.

The complexity bound is stated in η_{dec} as well. Part I's `thm: Upper bound successful iter BTR` gives

$$|\{k < T_\varepsilon \,:\, \rho_k \geq \eta_{\text{dec}}\}| \;\leq\; \frac{2\big(f(x_0) - \kappa_{\text{lbf}}\big)\big(L + \kappa_{\text{umh}}\big)}{\eta_{\text{dec}}\, \kappa_{\text{mdc}}\, \kappa_{\text{lbd}}}\, \varepsilon^{-2}. \tag{1}$$

The denominator carries η_{dec}, not η. The classical argument turns a successful iteration into a decrease of at least η times the predicted reduction and is vacuous at $\eta = 0$. Part I's argument turns an iteration with $\rho_k \geq \eta_{\text{dec}}$ into a decrease of at least η_{dec} times the predicted reduction. Such an iteration is accepted whenever $\eta_{\text{dec}} \geq \eta$, so the decrease is realised. The route is the structural conditions, and (1) is not vacuous at $\eta = 0$.

One discrepancy, which we report rather than resolve. The hypotheses of `thm: Upper bound successful iter BTR` open with Algorithm 1 applied with $\eta > 0$. The bound it proves and the proof it gives use η only through $\eta_{\text{dec}} \geq \eta$, which is vacuous at $\eta = 0$. Two readings are available. Under the first, the theorem is stated for $\eta > 0$ and says nothing at $\eta = 0$. Under the second, the hypothesis $\eta > 0$ is redundant and the conclusion holds whenever $\eta_{\text{dec}} > 0$. Choosing between them changes what Part I claims, so we choose neither. The author should settle it.

What $\eta = 0$ does cost. The upgrade from lim inf to lim is lost, and Part I says so. Its `th: firt-order TR CV` assumes $\eta > 0$ and concludes that $\liminf_{k\to\infty} \|g_k\| = 0$ implies $\lim_{k\to\infty} \|g_k\| = 0$. That theorem holds independently of the radius rule, so no condition on $\{\Delta_k\}$ recovers it. The remark that follows records the failure at $\eta = 0$. For iterations with $\rho_k \in (\eta, \eta_1)$ the algorithm can accept steps with arbitrarily small decrease, which admits accumulation points with non-zero gradient norm.

Consequence for this report. The band is a design freedom Part I permits and the framework of Curtis and Scheinberg forbids. It is not a practical gain bought against a lost complexity bound. It is a practical question asked inside a theory that already covers it, and the one guarantee at stake is the convergence of the whole sequence rather than of a subsequence. Every measurement below should be read against that.

2 What the solver does

A rejected step leaves the iterate in place. The next iteration reuses the gradient and pays one objective evaluation. An accepted step moves the iterate and pays a gradient as well. Lowering η converts rejected iterations into accepted ones, so it converts cheap iterations into expensive ones.

We confirmed this from the source rather than assuming it. In `src/Trust-region/common.jl` the acceptance test is `st.accepted = st.ρ >= p.η`, and the call `grad!(nlp, c.x, c.g)` sits inside the branch `if st.accepted`. The model update `update_model!` and the curvature estimate are inside the same branch. Outside the iteration the gradient is evaluated once, in `_init_state!`. The only other gradient call is in `_refresh_state!`, which the two sampled solvers use after a resample and which no run in this report reaches.

The gradient is therefore evaluated once per accepted iteration and never on a rejected one. An iteration count is not a work count here, and the two can move in opposite directions. We report function, gradient and Hessian-vector evaluations separately for that reason, and label iteration counts secondary wherever they appear.

3 Design

The sweep. For each rule we hold everything else fixed and sweep η over $\{0,\ \frac{1}{4}\eta_1,\ \frac{1}{2}\eta_1,\ \frac{3}{4}\eta_1,\ \eta_1\}$. The last is the classical coupling and is the baseline of every comparison.

Two strata. The first takes $\eta_1 = 0.1$, the value the campaign of Part III uses. The second takes $\eta_1 = 0.4$, because a band of width 0.1 in ρ may be too narrow to separate anything. Changing η_1 changes the rule's own behaviour, so the second stratum is a separate axis. We report the two separately and never pool them.

The rules. Part III carries two rosters. Its appendix table lists eleven radius rules by what the radius is tied to, which ratio selects the branch, whether the factor is fixed or adaptive, and where each is first stated. Its body table lists eleven and marks eight of them tested. The eight are R-delta, R-step, R-DFO(ω), R-grad($\omega, \overline{\mu}$), R-grad(ω), R-RTR, R-adaptive-step and R-adaptive-grad(ω). Those eight carry the primary tables. They are the package's `RULES`, held at one set of constants, $\gamma_1 = 0.25$, $\gamma_2 = 0.5$, $\gamma_3 = 2.0$.

The two rosters disagree on their eleventh entry, and the disagreement bears on the count, so we record it. The body table names R-delta-step, R-gRTR($\omega, \overline{\mu}$) and R-trunc-grad as the three untested rules. The appendix table names R-delta-step, R-gRTR($\omega, \overline{\mu}$) and R-adaptive-grad($\omega, \overline{\mu}$), and carries no entry for R-trunc-grad. Searching the source, R-trunc-grad has a macro in the preamble and one appearance in the body table, and no environment defines it anywhere. The author's own note in the source flags the same gap. We follow the appendix table, which defines every rule it names.

Three rules are therefore surveyed and untested, and all three are implemented. They are R-delta-step, R-adaptive-grad($\omega, \overline{\mu}$) and R-gRTR($\omega, \overline{\mu}$), carried by the package types `RDeltaStep`, `RAdaptiveGradCapped` and `RRTRGrad`. We report them in Section **??** and never inside the primary tables.

No package rule sits outside the surveyed taxonomy. Eleven rules are surveyed by the appendix table and eleven package types implement them. The second-order wrapper and its five aliases are not separate rules. They supply the criticality measure ω and forward the update unchanged.

Constructor refusals. `validate_thresholds` refuses $\eta_1 = 0$ for the step-driven rules R-step, R-adaptive-step and R-delta-step. This sweep varies η and holds η_1 at 0.1 or 0.4, so $\eta_1 = 0$ never arises and no refusal was expected. None occurred, and the archive records the empty list.

Counting the band. Occupancy is counted on accepted iterations as $\rho_k < \eta_1$, from the solver's own trace,

```
count(i -> accepted[i] && rho[i] < eta1, eachindex(accepted))
```

We do not count radius decreases. R-delta contracts by γ_2 on every mildly successful iteration, so a shrinking radius does not isolate the band. The count is zero by construction whenever $\eta = \eta_1$, and we report that column as the check that the instrument works.

The problem set. CUTEst, unconstrained, $2 \leq n \leq 200$, which realises 185 problems. This is the set Section 5 of Part III reports. The package's `default_problems()` falls back to eight analytic problems when the CUTEst query returns empty, which would produce a table that reads as a CUTEst benchmark without being one. The experiment file checks `HAS_CUTEST` and the realised count and stops rather than falling back. The realised list of names and dimensions is written to `tables/exp14_problem_set.txt` beside the results.

The model Hessian is exact and the subproblem solver is truncated conjugate gradient throughout. `ExactMS` is admissible on all 185 problems at $n \leq 200$, and we do not use it, so that no subsolver switch is confounded with the swept threshold.

Pairing. The comparison is η against η_1 within a rule on the same problem, so it is paired. A pair enters the log ratio only when both settings solve the problem. Problems solved by exactly one setting are named individually instead, because a log ratio over survivors would drop exactly the cases that matter. The tie band is $|\log(w_\eta/w_{\eta_1})| \leq 0.05$.

PLACEHOLDER-RESULTS